

Younsoo Park

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Education

The Pennsylvania State University

University Park, PA

Bachelor of Science in Computer Science Minor in Mathematics Expected Graduation: May 2027 Dean's List

Research Interests

Systems Security
Network Security
Distributed Systems
LLM Systems
AI Agent Memory
Federated Learning
TinyML for IoT Security
Adversarial Machine Learning
Edge AI Systems

Research Experience

Undergraduate Research Assistant Team Lead

Oct 2025 – Present

Reliable Federated TinyML Deployment for IoT Security (WIP, LCTES '26)

Penn State University and Elizabethtown College Supervisors: Dr. Suman Saha and Dr. Peilong Li

Research focus: Federated and adversarially robust TinyML systems for IoT security and resource-constrained edge-device deployment.

Key contributions:

- Designed a federated learning pipeline using FedAvgM, focal loss, and cosine learning-rate scheduling on CIC-IDS2017.
- Improved attack recall from 46.7
- Built a TinyML compression pipeline including BatchNorm folding, structured pruning, knowledge distillation, and INT8 QAT/PTQ.
- Achieved $12.28\times$ model compression and 74.5
- Evaluated adversarial robustness using PGD attacks under multiple compression configurations.

Publications and Manuscripts

Reliable Federated TinyML Deployment for IoT Security Work in Progress Target Venue: ACM LCTES 2026

Independent Study

AI Agent Memory Systems

Jul 2026 – Present

Advisor: Prof. Kiwan Maeng

- Surveying production AI agent memory systems including Mem0 and A-MEM.
- Investigating memory extraction, retrieval, safety, and lifecycle management for LLM-based agents.
- Applying memory architectures to production recommendation systems through a Memory-first Personalization framework.

Industry and Technical Experience

Levit Inc. (Shoppoort AI)

Apr 2026 – Present

Associate Problem Solver

- Worked directly with the CEO to define product direction, prioritize engineering initiatives, and translate customer problems into production AI systems for an LLM-powered shopping assistant.
- Built a production evaluation platform analyzing **49,807 conversations** and **80,124 query-level routing decisions**, achieving **97.7% evaluation coverage** through versioned routing audit and historical backfill pipelines.
- Designed versioned reclassification and historical backfill pipelines that reprocessed **47,231 conversations** spanning **100+ days** of production history, enabling continuous routing quality evaluation.
- Developed end-to-end experimentation infrastructure across frontend, backend, workers, and admin systems, improving recommendation completion from **62.8%** to **75.0% (+12.2 percentage points)**.
- Proposed a Memory-first Personalization architecture based on AI agent memory research and am currently developing a Mem0-based cross-category personalization proof of concept.

SIGNUM (Nittany AI Alliance – PIT-UN Funded)

Aug 2024 – Nov 2025

Co-founder of a \$3,750 PIT-UN funded AI platform for caregiver decision support.

- Built multi-source ETL pipelines integrating CMS, NPPES, and Google Places datasets into a DuckDB warehouse.
- Developed hospital-rating prediction models using Markov Transition analysis with confidence intervals.
- Designed and deployed domain-specific RAG systems using FAISS and AWS Bedrock.

Atom Tech Solutions LTD

Berkeley, CA

Software Engineering Intern

May 2022 – Aug 2022

- Built secure authentication systems using SQL/JWT authentication and encrypted credential storage.
- Implemented automated credential generation and authenticated onboarding workflows.

Technical Projects

XHIO – LinkedIn for Filmmakers

Apr 2026 – Present

- Built a creator networking platform featuring portfolios, video streaming, collaboration workflows, and school-based discovery using Next.js, Firebase, and Cloudflare Stream.
- Designed real-time messaging, creator profiles, media uploads, and production portfolio management.

ASME @ Penn State Website

Oct 2025 – Present

- Developed the official ASME Penn State website using React, TypeScript, Vite, and Firebase.
- Designed a multi-tier permission and approval system for organization management.

Hangukgwan (Family Business Web App)

2024 – Present

- Developed a multilingual full-stack restaurant web application using React, TypeScript, Node.js/Express, and MongoDB Atlas.
- Implemented JWT authentication, Google Maps API integration, and multilingual i18n support.

Teaching Experience

Lead Teaching Assistant, Math 230 (Multivariable Calculus) Penn State University Aug 2025 – Present

Responsibilities include assisting instruction, supporting students during office hours, and helping coordinate course activities.

Leadership and Service

THON Technology Committee Technology Captain (2027) Apr 2026 – Present

Penn State Advanced Vehicle Team (AVT) Project Manager Fall 2026 – Present

Association for Computing Machinery (ACM) Web and Design Director Apr 2026 – Present

Republic of Korea Air Force Squad Leader Sept 2022 – June 2024

Relevant Coursework

Computer Science

- CMPSC 131: Programming and Computation I
- CMPSC 132: Programming and Computation II
- CMPSC 221: Object Oriented Programming
- CMPSC 311: Introduction to Systems Programming
- CMPSC 360: Discrete Mathematics
- CMPSC 461: Programming Language Concepts
- CMPSC 464: Theory of Computation
- CMPSC 465: Data Structures and Algorithms

Computer Engineering

- CMPEN 270: Digital Design
- CMPEN 331: Computer Organization and Design

Awards and Honors

1st Place, ThinkNeuro LLC Project Competition	Apr 2026
1st Place, IEEE Student Chapter Battle Bots Competition	Apr 2026
PIT-UN Funding Award (\$3,750)	Apr 2025
3rd Place, ACM Pathfinder Robotics Challenge	Apr 2025
Republic of Korea Air Force Commendation Award	Aug 2023

Technical Skills

Programming Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL

Frameworks and Tools: React, Node.js, Express, Flask, MongoDB, Firebase, Git, AWS Bedrock, DuckDB

Machine Learning: TensorFlow, PyTorch, FAISS, SHAP, spaCy, Scikit-learn, Flower